

Program-9

Organize Kubernetes Pods Using Labels and Namespaces

Step 1: Create project directory

```
mkdir Program-9  
cd Program-9
```

Step 2: Create Namespace YAML file

Create a file named **namespaces.yaml**

```
nano namespaces.yaml
```

```
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

Step 3: Create Pods YAML file

Create a file named **pods.yaml**

```
nano pods.yaml
```

```
apiVersion: v1  
kind: Pod  
metadata:
```

```
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
      - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
      - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
```

```
    image: nginx
    ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
      - containerPort: 80
```

Step 4: Apply Namespaces

```
kubectl apply -f namespaces.yaml
```

Step 5: Apply Pods

```
kubectl apply -f pods.yaml
```

Step 6: Verify Pods by Namespace

```
kubectl get pods -n dev
kubectl get pods -n prod
```

Step 7: Check overall pod status

```
kubectl get pods
```

Step 8: Filter Pods by Label

List all frontend pods in dev namespace

```
kubectl get pods -n dev -l app=frontend
```

List all backend pods in prod namespace

```
kubectl get pods -n prod -l app=backend
```

```
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8: ~/demo9
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl apply -f namespaces.yaml
error: error validating "namespaces.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore these errors, turn validation off with --validate=false
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           17s
frontend-pod  1/1     Running   0           17s
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           80s
frontend-pod  1/1     Running   0           80s
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
kubia                               1/1     Running   3 (3m57s ago)    8d
nginx-deployment-6f9664446b-48ffd  1/1     Running   4 (6d17h ago)    20d
nginx-deployment-6f9664446b-nfcxw  1/1     Running   4 (6d17h ago)    20d
nodejs-pod                          1/1     Running   1 (3m57s ago)    6d18h
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           107s
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           2m1s
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~/demo9$
```